

Activity: Estimating Energy and Charging Needs for EV Evacuations

1. Summary

In this activity, participants use the “*Energy, Infrastructure, and Storage Needs for Electric Vehicle Evacuation Tool*” [1] to explore whether a community’s charging resources can support an evacuation with electric transportation.

Participants run a baseline scenario, change selected assumptions, and compare the estimated charging energy, required charger combinations, and backup-storage area. They then recommend preparedness improvements based on their findings.

2. Learning objectives

By the end of the activity, participants should be able to:

- Explain how evacuation distance, initial battery charge, and EV population affect charging-energy needs.
- Explain how evacuation time affects charging-infrastructure requirements.
- Recommend improvements to the energy infrastructure.

3. Preparation and materials

Organize participants into groups of four to six. Provide each group with:

- A computer with desktop Excel configured to run the workbook’s VBA macros.
- A working copy of excel file where the tool [1] is available .
- A worksheet for recording assumptions, results, and recommendations.

Before the activity, the instructor should test the introduce the worksheet and run the baseline simulation

Ask participants to read the Introduction worksheet and locate the inputs and charts in Inputs and Results. Modify only the designated inputs; leave the grey cells unchanged.

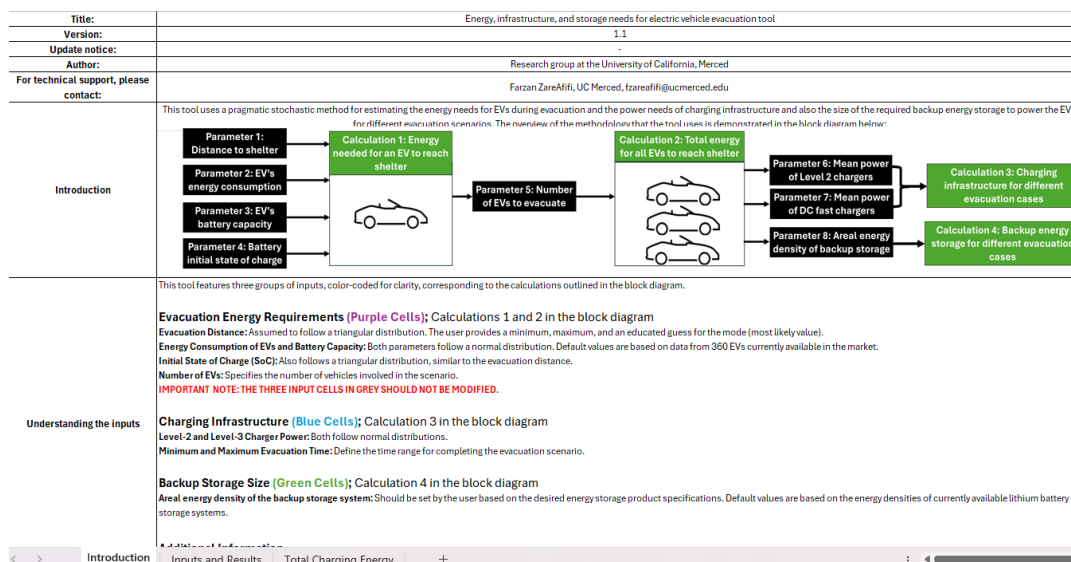


Figure 1. Screenshot of the excel worksheet to estimate charging needs during evacuations.

4. Baseline scenario

Consider a baseline scenario, where a community must evacuate 1,000 EVs because of an approaching wildfire. Residents travel different distances and begin with different battery states of charge.

Use the following baseline values, which match the attached workbook's default inputs:

Parameter	Baseline value
Evacuation distance: minimum / most likely / maximum	50 / 100 / 150 km
Initial state of charge: minimum / most likely / maximum	0 / 0.5 / 1
Number of EVs	1,000
Average EV energy consumption	0.188 kWh/km
Average EV battery capacity	74.4 kWh
Average Level-2 / Level-3 charger power	6 / 80 kW
Minimum / maximum evacuation time	4 / 8 hours
Minimum / maximum backup-storage areal energy density	160 / 570 kWh/m ²

Keep the workbook's standard deviations unchanged. State of charge is entered as a fraction: **0.5 means 50%**. The storage-density values are assumptions for this exercise; actual planning requires specifications for the selected storage system.

5. Suggested agenda

Stage	Time
Instructor introduces the scenario and tool	10 minutes
Groups predict outcomes and run the baseline	10 minutes
Groups investigate an assigned scenario	15 minutes
Groups interpret results and recommend improvements	15 minutes
Groups report findings and compare conclusions	10 minutes
Total	60 minutes

6. Activity instructions

Part A — Predict and run the baseline

Before running the simulation, ask participants:

1. Will every EV need charging before or during evacuation? Why?
2. Is total charging energy the same as the energy used by all EVs to travel to safety?
3. Would shortening the evacuation window change energy needs, charging-power needs, or both?

Run the baseline and record:

- The mean and standard deviation of total charging energy.
- One observation from the total and individual charging-energy distributions.
- The charger combinations shown for the minimum and maximum evacuation times.
- The backup-storage area shown for the relevant vehicle-count scenario.

Record chart labels and units so results are associated with the correct scenario.

Part B — Change one condition

Assign each group one scenario. Restore the baseline before making the assigned change.

Scenario	Input change
Longer evacuation routes	Change minimum / most likely / maximum distance to 80 / 150 / 220 km.
Lower initial battery charge	Change the most likely initial state of charge from 0.5 to 0.2, keeping its minimum and maximum unchanged.
Shorter evacuation window	Change minimum / maximum evacuation time from 4 / 8 to 2 / 4 hours.
More EVs evacuating	Change the EV population from 1,000 to 2,000.

Ask groups to predict the effect before running the tool, then record the same outputs as in Part A. Because the tool is stochastic, small differences between runs may reflect sampling variation.

Part C — Interpret and recommend

Each group should discuss:

- Which outputs changed most, and did the changes match your prediction?
- What do the charger combinations suggest about using Level-2 versus Level-3 charging?
- How could charging before an emergency affect preparedness?
- If grid power were unavailable, what do the storage results suggest about backup-energy needs?
- What additional information would be needed to turn these estimates into an operational plan?

Ask each group to recommend two preparedness improvements, explain how the results support them, and identify one implementation constraint.

7. Reporting and reflection

Each group gives a brief presentation covering its input change, main result, and recommendations. Collect a one-page worksheet comparing baseline and modified results, with at least one chart or screenshot.

Final remarks:

- Reflection question: “Which would you prioritize for this community: improving initial battery charge, adding charging capacity, or providing backup energy? Under what conditions would your choice change?”
- **Limitations** of the tool [1]: The tool provides estimates for planning discussions. Meeting an aggregate energy requirement does not by itself establish that everyone can evacuate successfully. Charger locations, queues, road congestion, grid capacity, equipment compatibility, and access for vulnerable residents also need consideration.

Reference

[1] ZareAfifi, F., de Castro, R., and Kurtz, S. (2024). *Planning for electric-vehicle evacuations: energy, infrastructure, and storage needs*. IEEE Vehicle Power and Propulsion Conference (VPPC). See the [UC Merced publication listing](#).

Participant Worksheet

Group members: _____

Assigned scenario: _____

1. Record your assumptions

Complete the modified-scenario column. Keep all other inputs at their baseline values.

Input	Baseline	Modified scenario
Number of EVs	1,000	
Minimum evacuation distance (km)	50	
Most likely evacuation distance (km)	100	
Maximum evacuation distance (km)	150	
Minimum initial state of charge	0	
Most likely initial state of charge	0.5	
Maximum initial state of charge	1	
Minimum evacuation time (hours)	4	
Maximum evacuation time (hours)	8	

2. Predict the effects before running the tool

For each output, write increase, decrease, or little or no change, and briefly explain your prediction.

Output	Predicted change	Reason
Mean total charging energy		
Variability of total charging energy		
Required charging capacity		
Backup-storage area for the same storage energy density		

3. Record the simulation results

Use the same units and matching scenario labels when comparing results.

Output	Baseline result	Modified result	Main observation
Mean total charging energy (MWh)			
Standard deviation of total charging energy (MWh)			
Observation from the individual EV charging-energy distribution			
Observation from the total charging-energy histogram			

Did the results match your predictions? Explain any differences.

4. Compare charging-infrastructure requirements

Choose a charger combination from each relevant curve. To make the comparison easier, use the same number of Level-2 chargers in all four rows and read the corresponding Level-3 requirement.

Record approximate chart values; round required charger counts upward when proposing an installation.

Simulation	Evacuation time used (hours)	Selected Level-2 charger count	Corresponding Level-3 charger count
Baseline — minimum time			
Baseline — maximum time			
Modified — minimum time			
Modified — maximum time			

What does this comparison show about evacuation time and charging requirements?

5. Compare backup-storage area

Read the storage-area chart at the same selected areal energy density for both simulations. Identify the vehicle-count curve used.

Item	Baseline	Modified scenario
Vehicle-count curve used		
Selected areal energy density (kWh/m ²)		
Estimated storage area (m ²)		

The plotted storage area is an estimate based on energy density. A complete installation may require additional space for equipment, access, and site requirements.

6. Develop preparedness recommendations

Recommend two improvements supported by your findings.

Proposed improvement	Supporting result	Implementation constraint or information needed
1.		
2.		

7. Prepare your group report

Question	Group response
What was your most important finding?	
Which preparedness improvement would you prioritize, and why?	
What important factor does this analysis leave unresolved?	

Question	Group response
What additional local data would improve the estimates?	